STIG Audit Basic Complete Check Reference

Every implemented IOS and IOS XE L2 and NDM check

This reference explains all 121 check definitions supplied in this Basic package: 22 IOS L2, 35 IOS NDM, 22 IOS XE L2, and 42 IOS XE NDM. Each entry gives the requirement, purpose, collected evidence, operator inputs, actual decision method, and outcome mapping. Use a V number to locate an entry with Word Find or the PDF search tool.

How the app determines a result

The selected platform and family load the corresponding YAML files from Application\data\checks. Each check declares read-only commands, conditions, status mappings, and comments. The app collects approved output over SSH, or loads the IOS-XE demonstration files in Sample outputs mode. The engine parses configuration, interfaces, neighbors, ACLs, and other fields as needed and evaluates the conditions with the selected profile. Basic writes new TXT and CKL reports as encrypted .sapenc files in the selected output folder. Open a protected TXT in Results; deliberately export a readable CKL for STIG Viewer and handle that unencrypted copy under site policy. Basic does not save an evidence archive or expose Pro history and review screens; retain the report and relevant approved device evidence through your assessment process.

Missing required command output returns the check's error status. An evaluation exception also returns that status. An ordinary true or false result maps to the configured pass or fail status listed for that entry. Some evaluators override those mappings for Error, Not_Reviewed, or conditional applicability. NotAFinding is a configured pass, Open a finding, Not_Applicable an applicability decision, and Not_Reviewed an unfinished assessment. The manual presets are not device verification. A pattern-only check can miss context, and no-candidate interface checks can pass; review the scope and actual evidence.

The nine manual entries describe local assumptions. Confirm these separately with the system owner and record the evidence and reviewer decision in the completed checklist. Official checklists can contain requirements beyond the implemented checks. In a blank CKL, unmatched requirements remain Not_Reviewed. Error and Skipped are written to CKL as Not_Reviewed.

Information to obtain from the site owner

All live runs need switch management addresses or hostnames, SSH credentials, and sometimes an enable secret. Those are connection inputs, not evidence that a security control passes. Each per-check Site input section identifies the exact profile values used. A field written as ntp_servers in the engine is stored under variables.ntp_servers in the YAML. Values shown below are the packaged baseline, not a recommendation for another site.

- unused_vlan: VLAN reserved for unused access ports. Ask the network owner for the approved number.
- native_vlan: Approved native VLAN number on trunks.
- management_vlan: Management SVI VLAN used to select the management address recorded in CKL asset metadata. The protected report filename identifies the selected scan target, platform, family, date, and run.
- dhcp_snooping.vlans: User VLAN numbers expected to have DHCP snooping configured.
- arp_inspection.vlans: User VLAN numbers expected to have Dynamic ARP Inspection configured.
- root_guard.upstream_switches: Exact upstream core/distribution Device IDs from show cdp neighbors detail. Short names and fully qualified names are matched. These links are exempt from Root Guard.
- management_access.acl_name: Name of the ACL applied inbound to VTY management lines.
- management_access.networks: Approved source networks as network_address and subnet_mask. Use an ordinary subnet mask, not an ACL wildcard mask.
- endpoint_authentication.radius_group: AAA RADIUS server group name as configured on the switch.
- endpoint_authentication.radius_servers: List of required radius server names; use the configuration names, not necessarily DNS hostnames.

- endpoint_authentication.radius_server_addresses: Mapping of each listed RADIUS configuration name to its expected IPv4 address.
- variables.logging_servers: Exact IP addresses or hostnames following logging host in the configuration. List at least two approved destinations.
- variables.ntp_servers: Exact IP addresses or hostnames following ntp server. List at least two approved sources.
- variables.supported_ios_versions: Exact approved IOS release strings from show version. The shipped IOS value is 15.2(7)E14; the organization must maintain the list.
- variables.supported_iosxe_versions: Exact approved IOS-XE release strings from show version. The shipped value is 17.12.08; the organization must maintain the list.
- variables.local_password_policy: Name of the aaa common-criteria policy referenced by local user configuration.
- variables.tacacs_group: TACACS+ server group name used by the default login method list.
- variables.max_concurrent_management_sessions: Organization-approved maximum concurrent management sessions, a positive integer.
- trunk_policy.additional_pruned_vlans: Additional VLAN numbers forbidden on trunks, where referenced by a check.

Important implementation details

- IOS-XE V-220554 currently allows both hmac-sha2-256 and md5 in the active YAML. Its descriptive comment mentions only HMAC-SHA2-256. The actual algorithm allowlist controls the result; an app pass is not independent FIPS validation.
- IOS V-220606 returns Open if no NTP authentication key exists or any MD5 key is present. A non-MD5 key with no MD5 key produces Not_Reviewed, requiring manual FIPS verification.
- IOS V-220583 through V-220585 return Not_Applicable without persistent logging; when enabled they require file privilege 15.
- IOS V-220618 is a preset pass assuming backups through Cisco Catalyst Center. IOS V-220619 is a preset Not Applicable for the supplied certificate-use assumption. Confirm each assumption at the tester's site.
- IOS-XE V-220528 requires the complete service timestamps log datetime msec localtime show-timezone year line. A missing year or an extra token after year fails the current expression. IOS V-220580 accepts trailing fields.
- Pattern descriptions summarize intent. Exact regular expressions, scopes, comments, and settings for all 121 entries are included in check_definitions.json and the editable YAML. If you change a check or profile, keep a copy with the assessment because this document describes the packaged version.

IOS L2 checks

22 implemented checks. File: ios_l2.yaml.

V-220623 CISC-L2-000020

The Cisco switch must uniquely identify and authenticate all network-connected endpoint devices before establishing any connection.

Purpose and required control: Require 802.1X and MAB on every physical access port and require all profile-defined RADIUS servers in the profile-defined AAA server group

Evidence read: show running-config, show interfaces status

Site input: endpoint_authentication.radius_group = "ISE-RADIUS". AAA RADIUS server group name as configured on the switch.

Site input: endpoint_authentication.radius_servers = ["RADIUS-SERVER-1", "RADIUS-SERVER-2"]. List of required radius server names; use the configuration names, not necessarily DNS hostnames.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "access"}}]}}

- Required on each interface: authentication port-control auto
- Required on each interface: dot1x pae authenticator
- Required on each interface: mab
- Required globally: RADIUS group {{ endpoint_authentication.radius_group }}
- Required globally for profile values: RADIUS server {{ item }} in group {{ endpoint_authentication.radius_group }} (profile field endpoint_authentication.radius_servers)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: All physical access ports have 802.1X and MAB, and all profile-defined RADIUS servers are configured in the required AAA group.

Configured failure explanation: One or more access-port authentication commands, the required AAA RADIUS group, or a profile-defined RADIUS server is missing.

Traceability: SV-220623r863275_rule | V3R2 (2026) | cat1 | interface_config_policy | ios_l2.yaml

V-220624 CISC-L2-000030

The Cisco switch must authenticate all VLAN Trunk Protocol (VTP) messages with a hash function using the most secured cryptographic algorithm available.

Purpose and required control: Looking to see if VTP status is off in "show vtp status"

Evidence read: show vtp status, show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Alternative: VTP operating mode is Off (read from show vtp status)
- Alternative: VTP password is configured (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: SV-220624r1188386_rule | V3R2 (2026) | cat2 | command_pattern_policy | ios_l2.yaml

V-220625 CISC-L2-000040

The Cisco switch must manage excess bandwidth to limit the effects of packet-flooding types of denial-of-service (DoS) attacks.

Purpose and required control: Require the global mls qos command in show running-config

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: Global mls qos is configured (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The global mls qos command is configured.

Configured failure explanation: The global mls qos command is missing.

Traceability: SV-220625r991847_rule | V3R2 (2026) | cat2 | command_pattern_policy | ios_l2.yaml

V-220629 CISC-L2-000090

The Cisco switch must have Root Guard enabled on all switch ports connecting to access layer switches.

Purpose and required control: Use CDP to identify connected switches. Exempt core/distribution neighbors containing 9500 or 9600, and require spanning-tree guard root on access-switch-facing ports.

Evidence read: show running-config, show cdp neighbors detail

Site input: root_guard.upstream_switches = ["CORE-DIST-01", "CORE-DIST-02"]. Exact upstream core/distribution Device IDs from show cdp neighbors detail. Short names and fully qualified names are matched. These links are exempt from Root Guard.

How it checks and verifies: The parser identifies switch neighbors from CDP. A configured upstream name exempts its local link; short names and fully qualified forms are compared. Any configured model token found in the neighbor Device ID or platform also exempts the link. Other switch-facing links require the exact Root Guard command on the local interface or its Port-channel. Missing CDP, missing exemption inputs, or an unresolved local interface can return Not_Reviewed. Missing Root Guard fails. When no unexempted switch neighbor remains, the test passes.

- upstream_profile_key: "root_guard.upstream_switches"
- required_string: "spanning-tree guard root"
- upstream_model_tokens: ["9500", "9600"]

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Root Guard is configured on all CDP-identified access-switch-facing interfaces.

Configured failure explanation: Root Guard is missing on one or more CDP-identified access-switch-facing interfaces.

Traceability: SV-220629r856223_rule | V3R2 (2026) | cat3 | root_guard_neighbor_policy | ios_l2.yaml

V-220630 CISC-L2-000100

The Cisco switch must have Bridge Protocol Data Unit (BPDU) Guard enabled on all user-facing or untrusted access switch ports.

Purpose and required control: Check if "spanning-tree bpduguard enable" is set globally, and if not, make sure it is applied to all access ports

Evidence read: show running-config, show interfaces status

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "access"}}]}}

- Required on each interface: spanning-tree bpduguard enable

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: SV-220630r856224_rule | V3R2 (2026) | cat2 | interface_config_policy | ios_l2.yaml

V-220631 CISC-L2-000110

The Cisco switch must have Spanning Tree Protocol (STP) Loop Guard enabled.

Purpose and required control: Looking for list of strings in "show run"

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: {"strings": ["spanning-tree loopguard default"], "command": "show running-config"} (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: SV-220631r856225_rule | V3R2 (2026) | cat2 | command_pattern_policy | ios_l2.yaml

V-220632 CISC-L2-000120

The Cisco switch must have Unknown Unicast Flood Blocking (UUFB) enabled.

Purpose and required control: Look through all access ports to ensure that "switchport block unicast" is configured on all access ports.

Evidence read: show running-config, show interfaces status

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "access"}}]}}

- Required on each interface: switchport block unicast

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: SV-220632r856226_rule | V3R2 (2026) | cat2 | interface_config_policy | ios_l2.yaml

V-220633 CISC-L2-000130

The Cisco switch must have DHCP snooping for all user VLANs to validate DHCP messages from untrusted sources.

Purpose and required control: Ensure global DHCP snooping and all profile-defined user VLANs are configured

Evidence read: show running-config

Site input: dhcp_snooping.vlans = [565]. User VLAN numbers expected to have DHCP snooping configured.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: global ip dhcp snooping enabled (read from show running-config)
- Required for every profile value: DHCP snooping VLAN {{ item }} (profile field dhcp_snooping.vlans; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: SV-220633r929007_rule | V3R2 (2026) | cat2 | command_pattern_policy | ios_l2.yaml

V-220634 CISC-L2-000140

The Cisco switch must have IP Source Guard enabled on all user-facing or untrusted access switch ports.

Purpose and required control: Look through all access ports to ensure that "ip verify source" is configured on all access ports.

Evidence read: show running-config, show interfaces status

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "access"}}]}}

- Required on each interface: ip verify source

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: SV-220634r929009_rule | V3R2 (2026) | cat2 | interface_config_policy | ios_l2.yaml

V-220635 CISC-L2-000150

The Cisco switch must have Dynamic Address Resolution Protocol (ARP) Inspection (DAI) enabled on all user VLANs.

Purpose and required control: Ensure Dynamic ARP Inspection is configured for all profile-defined user VLANs

Evidence read: show running-config

Site input: arp_inspection.vlans = [565]. User VLAN numbers expected to have Dynamic ARP Inspection configured.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required for every profile value: ARP inspection VLAN {{ item }} (profile field arp_inspection.vlans; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: SV-220635r929011_rule | V3R2 (2026) | cat2 | command_pattern_policy | ios_l2.yaml

V-220636 CISC-L2-000160

The Cisco switch must have Storm Control configured on all host-facing switchports.

Purpose and required control: Loop through all access ports to ensure multiple strings are applied

Evidence read: show running-config, show interfaces status

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure

causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "access"}}]}}

- Required on each interface: storm-control broadcast level

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: SV-220636r1205856_rule | V3R2 (2026) | cat3 | interface_config_policy | ios_l2.yaml

V-220637 CISC-L2-000170

The Cisco switch must have IGMP or MLD Snooping configured on all VLANs.

Purpose and required control: Ensure IGMP and IPv6 MLD snooping are not disabled globally or for individual VLANs

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Forbidden: IGMP snooping is disabled globally or for a VLAN (read from show running-config)
- Forbidden: IPv6 MLD snooping is disabled globally or for a VLAN (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: IGMP and IPv6 MLD snooping are not disabled globally or for individual VLANs.

Configured failure explanation: IGMP or IPv6 MLD snooping is disabled globally or for one or more VLANs.

Traceability: SV-220637r539671_rule | V3R2 (2026) | cat3 | command_pattern_policy | ios_l2.yaml

V-220638 CISC-L2-000180

The Cisco switch must implement Rapid Spanning Tree Protocol (STP) where VLANs span multiple switches with redundant links.

Purpose and required control: Looking for list of strings in "show run"

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Alternative: {"strings": ["spanning-tree mode rapid-pvst"], "command": "show running-config"} (read from show running-config)
- Alternative: {"strings": ["spanning-tree mode mst"], "command": "show running-config"} (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: SV-220638r539671_rule | V3R2 (2026) | cat2 | command_pattern_policy | ios_l2.yaml

V-220639 CISC-L2-000190

The Cisco switch must enable Unidirectional Link Detection (UDLD) to protect against one-way connections.

Purpose and required control: Looking for list of strings in "show run | inc udld"

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: {"command": "show running-config", "strings": ["udld enable"]} (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: SV-220639r1188389_rule | V3R2 (2026) | cat2 | command_pattern_policy | ios_l2.yaml

V-220640 CISC-L2-000200

The Cisco switch must have all trunk links enabled statically.

Purpose and required control: Ensure all switchports report Negotiation of Trunking as Off

Evidence read: show interfaces switchport | include Negotiation of Trunking

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: at least one switchport reports trunk negotiation Off (read from show interfaces switchport | include Negotiation of Trunking)
- Forbidden: switchport with trunk negotiation not set to Off (read from show interfaces switchport | include Negotiation of Trunking)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: All reported switchports have Negotiation of Trunking set to Off.

Configured failure explanation: One or more switchports have Negotiation of Trunking enabled or set to an unexpected value.

Traceability: SV-220640r1188390_rule | V3R2 (2026) | cat2 | command_pattern_policy | ios_l2.yaml

V-220641 CISC-L2-000210

The Cisco switch must have all disabled switch ports assigned to an unused VLAN.

Purpose and required control: Exempt 802.1X access ports; require all other disabled or disconnected access ports to be shut down and assigned to the unused VLAN

Evidence read: show running-config, show interfaces status

Site input: unused_vlan = 999. VLAN reserved for unused access ports. Ask the network owner for the approved number.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "access", "admin_state": "disabled_or_notconnect"}}]}}

- Interface exemption: 802.1X authentication configured
- Required on each interface: shutdown configured
- Required on each interface: unused VLAN {{ unused_vlan }} assigned

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: All non-exempt disabled or disconnected access ports are shut down and assigned to the unused VLAN.

Configured failure explanation: One or more non-exempt disabled or disconnected access ports are not shut down or are not assigned to the unused VLAN.

Traceability: SV-220641r991848_rule | V3R2 (2026) | cat2 | interface_config_policy | ios_l2.yaml

V-220642 CISC-L2-000220

The Cisco switch must not have the default VLAN assigned to any host-facing switch ports.

Purpose and required control: Loop through access ports to make sure none are assigned to VLAN 1

Evidence read: show running-config, show interfaces status

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "access"}}]}}

- Required on each interface: non-default access VLAN configured
- Forbidden on each interface: access VLAN 1

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: SV-220642r991849_rule | V3R2 (2026) | cat2 | interface_config_policy | ios_l2.yaml

V-220643 CISC-L2-000230

The Cisco switch must have the default VLAN pruned from all trunk ports that do not require it.

Purpose and required control: Loop through trunk ports to ensure all have VLAN 1 pruned

Evidence read: show running-config, show interfaces status

Site input: trunk_policy.additional_pruned_vlans = []. Additional VLAN numbers forbidden on trunks, where referenced by a check.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "trunk"}}]}}

- Required on each interface: explicit trunk allowed VLAN list
- Forbidden on each interface: VLAN 1 allowed on trunk
- Forbidden on each interface for profile values: profile-pruned VLAN {{ item }} allowed on trunk (profile field trunk_policy.additional_pruned_vlans)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: SV-220643r991850_rule | V3R2 (2026) | cat2 | interface_config_policy | ios_l2.yaml

V-220644 CISC-L2-000240

The Cisco switch must not use the default VLAN for management traffic.

Purpose and required control: Ensure VLAN 1 is not the management VLAN and has no IP address assigned

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Forbidden: interface Vlan1 has an IP address (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: SV-220644r991852_rule | V3R2 (2026) | cat2 | command_pattern_policy | ios_l2.yaml

V-220645 CISC-L2-000250

The Cisco switch must have all user-facing or untrusted ports configured as access switch ports.

Purpose and required control: Require switchport mode access on every physical copper Ethernet port; ignore SFP/fiber ports used as uplinks

Evidence read: show running-config, show interfaces status

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"media_type": {"regex": "BaseTX"}}}]}}

- Required on each interface: switchport mode access

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: All physical copper Ethernet ports are statically configured as access ports.

Configured failure explanation: One or more physical copper Ethernet ports are not statically configured as access ports.

Traceability: SV-220645r991853_rule | V3R2 (2026) | cat2 | interface_config_policy | ios_l2.yaml

V-220646 CISC-L2-000260

The Cisco switch must have the native VLAN assigned to an ID other than the default VLAN for all 802.1q trunk links.

Purpose and required control: Ensure every physical trunk has an explicitly configured native VLAN other than VLAN 1

Evidence read: show running-config, show interfaces status

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "trunk"}}]}}

- Required on each interface: non-default native VLAN configured
- Forbidden on each interface: native VLAN 1

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: SV-220646r991854_rule | V3R2 (2026) | cat2 | interface_config_policy | ios_l2.yaml

V-220647 CISC-L2-000270

The Cisco switch must not have any switchports assigned to the native VLAN.

Purpose and required control: Ensure no physical switchport is assigned as an access port in the profile-defined native VLAN

Evidence read: show interfaces status

Site input: native_vlan = 333. Approved native VLAN number on trunks.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical"}]}}

- require_candidates: true
- Forbidden parsed interface value: access VLAN matches native VLAN {{ native_vlan }} (profile field native_vlan)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: No physical switchports are assigned to native VLAN {{ native_vlan }}.

Configured failure explanation: One or more physical switchports are assigned to native VLAN {{ native_vlan }}.

Traceability: SV-220647r991855_rule | V3R2 (2026) | cat2 | interface_config_policy | ios_l2.yaml

IOS NDM checks

35 implemented checks. File: ios_ndm.yaml.

V-220570 CISC-ND-000010

The Cisco switch must be configured to limit the number of concurrent management sessions to an organization-defined number.

Purpose and required control: Validate session-limit under every VTY section, or count the available VTY lines when session-limit is unsupported

Evidence read: show running-config

Site input: variables.max_concurrent_management_sessions = 16. Organization-approved maximum concurrent management sessions, a positive integer.

How it checks and verifies: The maximum comes from the profile. If any VTY section has session-limit, all VTY sections must have a valid session-limit no greater than that maximum. Otherwise the app counts distinct configured VTY line numbers and compares the count to the maximum. Missing VTY sections or invalid input cannot establish compliance. This is a configured-capacity check, not a count of current logged-in users.

- command: "show running-config"
- maximum_sessions_profile_key: "max_concurrent_management_sessions"

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The number of concurrent management sessions does not exceed the organization-defined maximum of {{ max_concurrent_management_sessions }}.

Configured failure explanation: The VTY session limit or number of available VTY lines exceeds the organization-defined maximum of {{ max_concurrent_management_sessions }}.

Traceability: SV-220570r960735_rule | V3R8 (2026) | cat2 | vty_session_limit_policy | ios_ndm.yaml

V-220575 CISC-ND-000140

The Cisco switch must be configured to enforce approved authorizations for controlling the flow of management information within the device based on control policies.

Purpose and required control: Require the profile-defined management ACL inbound on every VTY section and permit only profile-approved management networks

Evidence read: show running-config

Site input: management_access.acl_name = "MANAGEMENT_NET". Name of the ACL applied inbound to VTY management lines.

Site input: management_access.networks = [{"network_address": "192.0.2.0", "subnet_mask": "255.255.255.0"}]. Approved source networks as network_address and subnet_mask. Use an ordinary subnet mask, not an ACL wildcard mask.

How it checks and verifies: Every parsed VTY section must contain access-class followed by the profile ACL name and in (optional vrf-also). The named standard or extended ACL must contain permits for all approved source networks and no permit source outside that set. The parser converts profile subnet masks to wildcard masks for comparison. Missing VTY sections or invalid/missing profile network data returns Error. Unparseable permit sources fail. The check compares source networks; it does not prove all aspects of traffic filtering.

Decision: Criteria met: NotAFinding. Criteria not met: Not_Reviewed. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Every VTY section uses the required inbound management ACL, which permits only the profile-approved management networks.

Configured failure explanation: A VTY section is missing the required inbound management ACL, or the ACL permits an unapproved source network.

Traceability: SV-220575r1137875_rule | V3R8 (2026) | cat2 | management_access_policy | ios_ndm.yaml

V-220576 CISC-ND-000150

The Cisco switch must be configured to enforce the limit of three consecutive invalid logon attempts, after which time it must lock out the user account from accessing the device for 15 minutes.

Purpose and required control: Ensure login blocking is configured for 900 seconds after three failed attempts within 120 seconds

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: login block-for 900 attempts 3 within 120 (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Login blocking is configured for 900 seconds after three failed attempts within 120 seconds.

Configured failure explanation: The required login block-for 900 attempts 3 within 120 command is missing.

Traceability: SV-220576r960840_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220577 CISC-ND-000160

The Cisco switch must be configured to display the Standard Mandatory DoD Notice and Consent Banner before granting access to the device.

Purpose and required control: Verify the login banner contains every mandatory DoD notice and consent clause

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app extracts complete login banner blocks and normalizes case and punctuation. One complete banner must contain all of the built-in DoD notice and consent clauses. Missing clauses fail. This is a text comparison; it does not open an interactive login session to verify presentation.

- command: "show running-config"
- banner_type: "login"

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The configured login banner contains all mandatory Standard DoD Notice and Consent clauses.

Configured failure explanation: The login banner is missing or does not contain all mandatory Standard DoD Notice and Consent clauses.

Traceability: SV-220577r960843_rule | V3R8 (2026) | cat2 | dod_banner_policy | ios_ndm.yaml

V-220578 CISC-ND-000210

The Cisco device must be configured to audit all administrator activity.

Purpose and required control: Ensure administrator configuration changes generate informational logging messages

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: logging userinfo (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Administrator activity logging is enabled with logging userinfo.

Configured failure explanation: The logging userinfo command is missing.

Traceability: SV-220578r1205849_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220580 CISC-ND-000280

The Cisco switch must produce audit records containing information to establish when (date and time) the events occurred.

Purpose and required control: Require service timestamps log datetime msec localtime, allowing additional options afterward

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: service timestamps log datetime msec localtime with optional additional settings (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Log timestamps use millisecond local date and time, with optional additional settings.

Configured failure explanation: The service timestamps log datetime msec localtime command is missing.

Traceability: SV-220580r960894_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220581 CISC-ND-000290

The Cisco switch must produce audit records containing information to establish where the events occurred.

Purpose and required control: Require log-input on every deny statement in each interface-bound IPv4 ACL

Evidence read: show running-config, show ip access-lists

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser reads ACL entries in the scope specified below and requires selected deny entries to carry logging. Missing logging on a selected deny fails. Review the exact scope: this does not require logging on ACL entries outside it or prove that log messages reach a server.

- deny_statements: {"interface_bound_only": true, "include_standard_acls": true, "include_extended_acls": true, "include_ipv6_acls": false, "require_keyword": "log-input"}

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Every deny statement in each interface-bound IPv4 ACL includes log-input.

Configured failure explanation: One or more deny statements in an interface-bound IPv4 ACL are missing log-input.

Traceability: SV-220581r960897_rule | V3R8 (2026) | cat2 | acl_deny_logging_policy | ios_ndm.yaml

V-220583 CISC-ND-000380

The Cisco switch must be configured to protect audit information from unauthorized modification.

Purpose and required control: If logging persistent is absent, mark Not Applicable. If enabled, require file privilege 15 and no other file privilege value.

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: Empty running configuration returns Error. If logging persistent is absent, this pack assigns Not_Applicable. If present, the app searches the running configuration for file privilege values: at least one must exist and all found values must be 15. That passes; a missing value or any other value produces Open. This does not test file access by logging in with different users.

Decision: No persistent logging: Not_Applicable. Enabled with all found file privilege values equal to 15: NotAFinding. Enabled with absent/other values: Open. Empty/missing output: Error.

Traceability: SV-220583r960933_rule | V3R8 (2026) | cat2 | persistent_logging_privilege_policy | ios_ndm.yaml

V-220584 CISC-ND-000390

The Cisco switch must be configured to protect audit information from unauthorized deletion.

Purpose and required control: If logging persistent is absent, mark Not Applicable. If enabled, require file privilege 15 and no other file privilege value.

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: Empty running configuration returns Error. If logging persistent is absent, this pack assigns Not_Applicable. If present, the app searches the running configuration for file privilege values: at least one must exist and all found values must be 15. That passes; a missing value or any other value produces Open. This does not test file access by logging in with different users.

Decision: No persistent logging: Not_Applicable. Enabled with all found file privilege values equal to 15: NotAFinding. Enabled with absent/other values: Open. Empty/missing output: Error.

Traceability: SV-220584r960936_rule | V3R8 (2026) | cat2 | persistent_logging_privilege_policy | ios_ndm.yaml

V-220585 CISC-ND-000460

The Cisco switch must be configured to limit privileges to change the software resident within software libraries.

Purpose and required control: If logging persistent is absent, mark Not Applicable. If enabled, require file privilege 15 and no other file privilege value.

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: Empty running configuration returns Error. If logging persistent is absent, this pack assigns Not_Applicable. If present, the app searches the running configuration for file privilege values: at least one must exist and all found values must be 15. That passes; a missing value or any other value produces Open. This does not test file access by logging in with different users.

Decision: No persistent logging: Not_Applicable. Enabled with all found file privilege values equal to 15: NotAFinding. Enabled with absent/other values: Open. Empty/missing output: Error.

Traceability: SV-220585r960960_rule | V3R8 (2026) | cat2 | persistent_logging_privilege_policy | ios_ndm.yaml

V-220586 CISC-ND-000470

The Cisco switch must be configured to prohibit the use of all unnecessary and non-secure functions and services.

Purpose and required control: Ensure prohibited and unnecessary services are not enabled in the running configuration

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Forbidden: prohibited or unnecessary service is enabled (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: None of the prohibited or unnecessary services are enabled.

Configured failure explanation: One or more prohibited or unnecessary services are enabled.

Traceability: SV-220586r1043177_rule | V3R8 (2026) | cat1 | command_pattern_policy | ios_ndm.yaml

V-220587 CISC-ND-000490

The Cisco switch must be configured with only one local account to be used as the account of last resort in the event the authentication server is unavailable.

Purpose and required control: Require exactly one local username using the profile-defined common-criteria password policy and require the profile-defined TACACS group before local in the default login authentication order

Evidence read: show running-config

Site input: variables.local_password_policy = "PASSWORD_POLICY". Name of the aaa common-criteria policy referenced by local user configuration.

Site input: variables.tacacs_group = "ISE-TACACS". TACACS+ server group name used by the default login method list.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: one local username account is configured (read from show running-config)
- Forbidden: multiple local username accounts are configured (read from show running-config)
- Required for every profile value: local username using common-criteria policy {{ item }} (profile field local_password_policy; read from show running-config)
- Required for every profile value: default login authentication uses group {{ item }} before local (profile field tacacs_group; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Exactly one local account uses the required common-criteria password policy, and the required TACACS group precedes local in the default login authentication order.

Configured failure explanation: The local-account count, common-criteria password policy, or default AAA login authentication order is incorrect.

Traceability: SV-220587r1051115_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220589 CISC-ND-000550

The Cisco switch must be configured to enforce a minimum 15-character password length.

Purpose and required control: Require min-length 15 under the profile-defined AAA common-criteria password policy

Evidence read: show running-config

Site input: variables.local_password_policy = "PASSWORD_POLICY". Name of the aaa common-criteria policy referenced by local user configuration.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required for every profile value: min-length 15 under common-criteria policy {{ item }} (profile field local_password_policy; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The required minimum password length is configured under the profile-defined common-criteria policy.

Configured failure explanation: The profile-defined common-criteria policy is missing min-length 15.

Traceability: SV-220589r1015280_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220590 CISC-ND-000570

The Cisco switch must be configured to enforce password complexity by requiring that at least one uppercase character be used.

Purpose and required control: Require upper-case 1 under the profile-defined AAA common-criteria password policy

Evidence read: show running-config

Site input: variables.local_password_policy = "PASSWORD_POLICY". Name of the aaa common-criteria policy referenced by local user configuration.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required for every profile value: upper-case 1 under common-criteria policy {{ item }} (profile field local_password_policy; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The uppercase-character requirement is configured under the profile-defined common-criteria policy.

Configured failure explanation: The profile-defined common-criteria policy is missing upper-case 1.

Traceability: SV-220590r1015281_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220591 CISC-ND-000580

The Cisco switch must be configured to enforce password complexity by requiring that at least one lowercase character be used.

Purpose and required control: Require lower-case 1 under the profile-defined AAA common-criteria password policy

Evidence read: show running-config

Site input: variables.local_password_policy = "PASSWORD_POLICY". Name of the aaa common-criteria policy referenced by local user configuration.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required for every profile value: lower-case 1 under common-criteria policy {{ item }} (profile field local_password_policy; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The lowercase-character requirement is configured under the profile-defined common-criteria policy.

Configured failure explanation: The profile-defined common-criteria policy is missing lower-case 1.

Traceability: SV-220591r1015282_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220592 CISC-ND-000590

The Cisco switch must be configured to enforce password complexity by requiring that at least one numeric character be used.

Purpose and required control: Require numeric-count 1 under the profile-defined AAA common-criteria password policy

Evidence read: show running-config

Site input: variables.local_password_policy = "PASSWORD_POLICY". Name of the aaa common-criteria policy referenced by local user configuration.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required for every profile value: numeric-count 1 under common-criteria policy {{ item }} (profile field local_password_policy; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The numeric-character requirement is configured under the profile-defined common-criteria policy.

Configured failure explanation: The profile-defined common-criteria policy is missing numeric-count 1.

Traceability: SV-220592r1015283_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220593 CISC-ND-000600

The Cisco switch must be configured to enforce password complexity by requiring that at least one special character be used.

Purpose and required control: Require special-case 1 under the profile-defined AAA common-criteria password policy

Evidence read: show running-config

Site input: variables.local_password_policy = "PASSWORD_POLICY". Name of the aaa common-criteria policy referenced by local user configuration.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required for every profile value: special-case 1 under common-criteria policy {{ item }} (profile field local_password_policy; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The special-character requirement is configured under the profile-defined common-criteria policy.

Configured failure explanation: The profile-defined common-criteria policy is missing special-case 1.

Traceability: SV-220593r1015284_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220594 CISC-ND-000610

The Cisco switch must be configured to require that when a password is changed, the characters are changed in at least eight of the positions within the password.

Purpose and required control: Require char-changes 8 under the profile-defined AAA common-criteria password policy

Evidence read: show running-config

Site input: variables.local_password_policy = "PASSWORD_POLICY". Name of the aaa common-criteria policy referenced by local user configuration.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required for every profile value: char-changes 8 under common-criteria policy {{ item }} (profile field local_password_policy; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The required password-position changes are configured under the profile-defined common-criteria policy.

Configured failure explanation: The profile-defined common-criteria policy is missing char-changes 8.

Traceability: SV-220594r1043189_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220595 CISC-ND-000620

The Cisco switch must only store cryptographic representations of passwords.

Purpose and required control: Ensure service password-encryption is enabled

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: service password-encryption enabled (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Service password encryption is enabled.

Configured failure explanation: The service password-encryption command is missing.

Traceability: SV-220595r1015286_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220596 CISC-ND-000720

The Cisco switch must be configured to terminate all network connections associated with device management after five minutes of inactivity.

Purpose and required control: Require session-timeout 5 or exec-timeout 5 0 under every VTY line section

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: at least one VTY line section is configured (read from show running-config)
- Forbidden: VTY line section missing an approved five-minute inactivity timeout (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Every VTY line section has a five-minute inactivity timeout configured.

Configured failure explanation: One or more VTY line sections are missing session-timeout 5 or exec-timeout 5 0.

Traceability: SV-220596r961068_rule | V3R8 (2026) | cat1 | command_pattern_policy | ios_ndm.yaml

V-220599 CISC-ND-000980

The Cisco switch must be configured to allocate audit record storage capacity in accordance with organization-defined audit record storage requirements.

Purpose and required control: Ensure logging buffered specifies a numeric buffer size and a valid logging severity level

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: logging buffered with numeric size and valid severity level (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Logging buffered is configured with a numeric buffer size and valid severity level.

Configured failure explanation: A valid logging buffered size and severity level are not configured.

Traceability: SV-220599r961392_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220600 CISC-ND-001000

The Cisco switch must be configured to generate an alert for all audit failure events.

Purpose and required control: Ensure remote logging uses a valid named trap severity level

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: logging trap with valid named severity level (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Remote logging is configured with a valid named trap severity level.

Configured failure explanation: A valid named logging trap severity level is not configured.

Traceability: SV-220600r991868_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220601 CISC-ND-001030

The Cisco switch must be configured to synchronize its clock with the primary and secondary time sources using redundant authoritative time sources.

Purpose and required control: Require at least two NTP server commands and every NTP server defined in the selected profile

Evidence read: show running-config

Site input: variables.ntp_servers = ["192.0.2.23", "192.0.2.22"]. Exact IP addresses or hostnames following ntp server. List at least two approved sources.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: at least two NTP server commands (read from show running-config)
- Required for every profile value: profile-defined NTP server {{ item }} (profile field ntp_servers; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: At least two profile-defined authoritative NTP servers are configured.

Configured failure explanation: Fewer than two NTP servers are configured or a profile-defined NTP server is missing.

Traceability: SV-220601r1015287_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220604 CISC-ND-001130

The Cisco switch must be configured to authenticate SNMP messages using a FIPS-validated Keyed-Hash Message Authentication Code (HMAC).

Purpose and required control: Require at least one SNMPv3 user using SHA or a supported SHA-2 authentication protocol

Evidence read: show snmp user

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: FIPS-acceptable SNMPv3 HMAC authentication protocol (read from show snmp user)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: At least one SNMPv3 user uses a FIPS-acceptable SHA-based HMAC authentication protocol.

Configured failure explanation: No SNMPv3 user using an approved SHA-based HMAC authentication protocol was found.

Traceability: SV-220604r961506_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220605 CISC-ND-001140

The Cisco switch must be configured to encrypt SNMP messages using a FIPS 140-2 approved algorithm.

Purpose and required control: Require at least one SNMPv3 user using AES-128, AES-192, or AES-256 privacy

Evidence read: show snmp user

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: FIPS-approved AES SNMPv3 privacy protocol (read from show snmp user)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: At least one SNMPv3 user uses a FIPS-approved AES privacy protocol.

Configured failure explanation: No SNMPv3 user using AES-128, AES-192, or AES-256 privacy was found.

Traceability: SV-220605r961506_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220606 CISC-ND-001150

The Cisco switch must be configured to authenticate Network Time Protocol (NTP) sources using authentication with FIPS-compliant algorithms.

Purpose and required control: Flag MD5 NTP authentication keys as Open; review any other configured algorithm for FIPS compliance

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: an NTP authentication key is configured (read from show running-config)
- Forbidden: an MD5 NTP authentication key is configured (read from show running-config)

Decision: Criteria met: Not_Reviewed. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: An NTP authentication key uses a non-MD5 algorithm; verify FIPS compliance manually.

Configured failure explanation: An MD5 NTP authentication key is configured, or no NTP authentication key was found. This is an open finding.

Traceability: SV-220606r1156814_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220607 CISC-ND-001200

The Cisco switch must be configured to use FIPS-validated Keyed-Hash Message Authentication Code (HMAC) to protect the integrity of remote maintenance sessions.

Purpose and required control: Require the approved SSH HMAC algorithms in the configured order

Evidence read: show running-config | include ssh

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: approved SSH HMAC algorithm list (read from show running-config | include ssh)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: SSH uses the approved HMAC-SHA2 integrity algorithms in the required order.

Configured failure explanation: The required SSH HMAC algorithm list is missing or incorrect.

Traceability: SV-220607r1205843_rule | V3R8 (2026) | cat1 | command_pattern_policy | ios_ndm.yaml

V-220608 CISC-ND-001210

The Cisco switch must be configured to implement cryptographic mechanisms to protect the confidentiality of remote maintenance sessions.

Purpose and required control: Require the approved SSH AES-CTR encryption algorithms in the configured order

Evidence read: show running-config | include ssh

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: approved SSH AES-CTR encryption algorithm list (read from show running-config | include ssh)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: SSH uses the approved AES-CTR encryption algorithms in the required order.

Configured failure explanation: The required SSH AES-CTR encryption algorithm list is missing or incorrect.

Traceability: SV-220608r961557_rule | V3R8 (2026) | cat1 | command_pattern_policy | ios_ndm.yaml

V-220609 CISC-ND-001220

The Cisco switch must be configured to protect against known types of denial-of-service (DoS) attacks by employing organization-defined security safeguards.

Purpose and required control: C-22324r1007810_chk
Review the Cisco switch configuration to verify that it protects against known types of DoS attacks by employing organization-defined security safeguards. Note: Cisco IOS Version 15 and newer could use the command "mls qos copp protocol" to meet the intent of this requirement. This can be checked using command "show mls qos copp

Evidence read: No command; manual preset.

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The engine does not independently test the device for this item. It returns the configured fail status as a preset, even when that status is NotAFinding or Not_Applicable. The assessor must obtain the organizational or device evidence described below and make the final decision.

Decision and required manual verification: Current preset: Open. Manual IOS guide review required; no automated compliance verdict.

Traceability: SV-220609r1056195_rule | V3R8 (2026) | cat2 | manual_review | ios_ndm.yaml

V-220612 CISC-ND-001260

The Cisco switch must be configured to generate audit records when successful/unsuccessful logon attempts occur.

Purpose and required control: Ensure successful and failed login attempts are logged

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: login on-success log (read from show running-config)
- Required: login on-failure log (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Successful and failed login attempts are configured for logging.

Configured failure explanation: Login success logging, login failure logging, or both are missing.

Traceability: SV-220612r961824_rule | V3R8 (2026) | cat2 | command_pattern_policy | ios_ndm.yaml

V-220617 CISC-ND-001370

The Cisco switch must be configured to use at least two authentication servers to authenticate users prior to granting administrative access.

Purpose and required control: Require at least two profile-defined RADIUS server blocks with the correct address, authentication port, accounting port, key, and AAA group membership

Evidence read: show running-config

Site input: endpoint_authentication.radius_group = "ISE-RADIUS". AAA RADIUS server group name as configured on the switch.

Site input: endpoint_authentication.radius_server_addresses = {"RADIUS-SERVER-1": "192.0.2.18", "RADIUS-SERVER-2": "192.0.2.19"}. Mapping of each listed RADIUS configuration name to its expected IPv4 address.

Site input: endpoint_authentication.radius_servers = ["RADIUS-SERVER-1", "RADIUS-SERVER-2"]. List of required radius server names; use the configuration names, not necessarily DNS hostnames.

How it checks and verifies: The profile must define at least two unique RADIUS server names and an IPv4 mapping for each. Each corresponding radius server section must use the mapped address, authentication port 1812, accounting port 1813, and a key command. The named AAA RADIUS group must contain each required server name. Missing or mismatched configuration fails. The app does not contact the RADIUS servers or test a login.

- radius_group_profile_key: "endpoint_authentication.radius_group"
- radius_servers_profile_key: "endpoint_authentication.radius_servers"
- radius_addresses_profile_key: "endpoint_authentication.radius_server_addresses"
- auth_port: 1812
- acct_port: 1813

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: At least two profile-defined RADIUS servers have the required addresses, ports, keys, and AAA group membership.

Configured failure explanation: One or more required RADIUS server definitions, addresses, ports, keys, or AAA group memberships are missing or incorrect.

Traceability: SV-220617r1205840_rule | V3R8 (2026) | cat1 | radius_server_policy | ios_ndm.yaml

V-220618 CISC-ND-001410

The Cisco switch must be configured to support organizational requirements to conduct backups of the configuration when changes occur.

Purpose and required control: Temporarily mark NotAFinding because configuration backups are performed through Cisco Catalyst Center

Evidence read: No command; manual preset.

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The engine does not independently test the device for this item. It returns the configured fail status as a preset, even when that status is NotAFinding or Not_Applicable. The assessor must obtain the organizational or device evidence described below and make the final decision.

Decision and required manual verification: Current preset: NotAFinding. Configuration backups are performed through Cisco Catalyst Center.

Traceability: SV-220618r1069531_rule | V3R8 (2026) | cat2 | manual_review | ios_ndm.yaml

V-220619 CISC-ND-001440

The Cisco switch must be configured to obtain its public key certificates from an appropriate certificate policy through an approved service provider.

Purpose and required control: Temporarily Not Applicable: only non-removable Cisco default certificates are present; revisit certificate applicability later

Evidence read: No command; manual preset.

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The engine does not independently test the device for this item. It returns the configured fail status as a preset, even when that status is NotAFinding or Not_Applicable. The assessor must obtain the organizational or device evidence described below and make the final decision.

Decision and required manual verification: Current preset: Not_Applicable. Not Applicable for now. These switches do not use organization-managed certificates; Cisco default certificates remain installed and cannot be deleted. Revisit this determination later.

Traceability: SV-220619r991871_rule | V3R8 (2026) | cat2 | manual_review | ios_ndm.yaml

V-220620 CISC-ND-001450

The Cisco switch must be configured to send log data to at least two central log servers for the purpose of forwarding alerts to the administrators and the information system security officer (ISSO).

Purpose and required control: Require at least two logging host commands and every logging server defined in the selected profile

Evidence read: show running-config

Site input: variables.logging_servers = ["192.0.2.5", "192.0.2.6"]. Exact IP addresses or hostnames following logging host in the configuration. List at least two approved destinations.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: at least two logging host commands (read from show running-config)
- Required for every profile value: profile-defined logging server {{ item }} (profile field logging_servers; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: At least two profile-defined central logging servers are configured.

Configured failure explanation: Fewer than two logging hosts are configured or a profile-defined logging server is missing.

Traceability: SV-220620r1137890_rule | V3R8 (2026) | cat1 | command_pattern_policy | ios_ndm.yaml

V-220621 CISC-ND-001470

The Cisco switch must be running an IOS release that is currently supported by Cisco Systems.

Purpose and required control: Match the installed IOS release against the profile-defined approved version

Evidence read: show version

Site input: variables.supported_ios_versions = ["15.2(7)E14"]. Exact approved IOS release strings from show version. The shipped IOS value is 15.2(7)E14; the organization must maintain the list.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Alternative from profile: profile-approved IOS version {{ item }} (profile field supported_ios_versions; read from show version)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The installed IOS release matches the profile-defined approved version.

Configured failure explanation: The installed IOS release is not in the profile-defined approved list.

Traceability: SV-220621r1207744_rule | V3R8 (2026) | cat1 | command_pattern_policy | ios_ndm.yaml

IOSXE L2 checks

22 implemented checks. File: iosxe_l2.yaml.

V-220649 CISC-L2-000020

The Cisco switch must uniquely identify and authenticate all network-connected endpoint devices before establishing any connection.

Purpose and required control: Require 802.1X and MAB on every physical access port and require all profile-defined RADIUS servers in the profile-defined AAA server group

Evidence read: show running-config, show interfaces status

Site input: endpoint_authentication.radius_group = "ISE-RADIUS". AAA RADIUS server group name as configured on the switch.

Site input: endpoint_authentication.radius_servers = ["RADIUS-SERVER-1", "RADIUS-SERVER-2"]. List of required radius server names; use the configuration names, not necessarily DNS hostnames.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "access"}}]}}

- Required on each interface: authentication port-control auto
- Required on each interface: dot1x pae authenticator
- Required on each interface: mab
- Required globally: RADIUS group {{ endpoint_authentication.radius_group }}

- Required globally for profile values: RADIUS server {{ item }} in group {{ endpoint_authentication.radius_group }} (profile field endpoint_authentication.radius_servers)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: All physical access ports have 802.1X and MAB, and all profile-defined RADIUS servers are configured in the required AAA group.

Configured failure explanation: One or more access-port authentication commands, the required AAA RADIUS group, or a profile-defined RADIUS server is missing.

Traceability: cat1 | interface_config_policy | iosxe_l2.yaml

V-220650 CISC-L2-000030

The Cisco switch must authenticate all VLAN Trunk Protocol (VTP) messages with a hash function using the most secured cryptographic algorithm available.

Purpose and required control: Looking to see if VTP status is off in "show vtp status"

Evidence read: show vtp status, show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Alternative: VTP operating mode is Off (read from show vtp status)
- Alternative: VTP password is configured (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: cat2 | command_pattern_policy | iosxe_l2.yaml

V-220651 CISC-L2-000040

The Cisco switch must manage excess bandwidth to limit the effects of packet flooding types of denial-of-service (DoS) attacks.

Purpose and required control: QoS is not currently configured; retain as Open until the site QoS policy is implemented and this check is automated

Evidence read: No command; manual preset.

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The engine does not independently test the device for this item. It returns the configured fail status as a preset, even when that status is NotAFinding or Not_Applicable. The assessor must obtain the organizational or device evidence described below and make the final decision.

Decision and required manual verification: Current preset: Open. Open pending QoS implementation. Configure and validate the site QoS policy, then replace this placeholder with an automated check.

Traceability: cat2 | manual_review | iosxe_l2.yaml

V-220655 CISC-L2-000090

The Cisco switch must have Root Guard enabled on all switch ports connecting to access layer switches.

Purpose and required control: Use CDP to identify directly connected switches. Exempt configured core/distribution neighbors and require "spanning-tree guard root" on all other switch-facing interfaces.

Evidence read: show running-config, show cdp neighbors detail

Site input: root_guard.upstream_switches = ["CORE-DIST-01"]. Exact upstream core/distribution Device IDs from show cdp neighbors detail. Short names and fully qualified names are matched. These links are exempt from Root Guard.

How it checks and verifies: The parser identifies switch neighbors from CDP. A configured upstream name exempts its local link; short names and fully qualified forms are compared. Any configured model token found in the neighbor Device ID or platform also exempts the link. Other switch-facing links require the exact Root Guard command on the local interface or its Port-channel. Missing CDP, missing exemption inputs, or an unresolved local interface can return Not_Reviewed. Missing Root Guard fails. When no unexempted switch neighbor remains, the test passes.

- upstream_profile_key: "root_guard.upstream_switches"
- required_string: "spanning-tree guard root"

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Root Guard is configured on all CDP-identified access-switch-facing interfaces.

Configured failure explanation: Root Guard is missing on one or more CDP-identified access-switch-facing interfaces.

Traceability: cat3 | root_guard_neighbor_policy | iosxe_l2.yaml

V-220656 CISC-L2-000100

The Cisco switch must have BPDU Guard enabled on all user-facing or untrusted access switch ports.

Purpose and required control: Check if "spanning-tree bpduguard enable" is set globally, and if not, make sure it is applied to all access ports

Evidence read: show running-config, show interfaces status

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "access"}}]}}

- Required on each interface: spanning-tree bpduguard enable

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: cat2 | interface_config_policy | iosxe_l2.yaml

V-220657 CISC-L2-000110

The Cisco switch must have STP Loop Guard enabled.

Purpose and required control: Looking for list of strings in "show run"

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: {"strings": ["spanning-tree loopguard default"], "command": "show running-config"} (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: cat2 | command_pattern_policy | iosxe_l2.yaml

V-220658 CISC-L2-000120

The Cisco switch must have Unknown Unicast Flood Blocking (UUFB) enabled.

Purpose and required control: Look through all access ports to ensure that "switchport block unicast" is configured on all access ports.

Evidence read: show running-config, show interfaces status

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "access"}}]}}

- Required on each interface: switchport block unicast

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: cat2 | interface_config_policy | iosxe_l2.yaml

V-220659 CISC-L2-000130

The Cisco switch must have DHCP snooping for all user VLANs to validate DHCP messages from untrusted sources.

Purpose and required control: Ensure global DHCP snooping and all profile-defined user VLANs are configured

Evidence read: show running-config

Site input: dhcp_snooping.vlans = [565]. User VLAN numbers expected to have DHCP snooping configured.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile

lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: global ip dhcp snooping enabled (read from show running-config)
- Required for every profile value: DHCP snooping VLAN {{ item }} (profile field dhcp_snooping.vlans; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: cat2 | command_pattern_policy | iosxe_l2.yaml

V-220660 CISC-L2-000140

The Cisco switch must have IP Source Guard enabled on all user-facing or untrusted access switch ports.

Purpose and required control: Look through all access ports to ensure that "ip verify source" is configured on all access ports.

Evidence read: show running-config, show interfaces status

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "access"}}]}}

- Required on each interface: ip verify source

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: cat2 | interface_config_policy | iosxe_l2.yaml

V-220661 CISC-L2-000150

The Cisco switch must have Dynamic Address Resolution Protocol (ARP) Inspection (DAI) enabled on all user VLANs.

Purpose and required control: Ensure Dynamic ARP Inspection is configured for all profile-defined user VLANs

Evidence read: show running-config

Site input: arp_inspection.vlans = [565]. User VLAN numbers expected to have Dynamic ARP Inspection configured.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required for every profile value: ARP inspection VLAN {{ item }} (profile field arp_inspection.vlans; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: cat2 | command_pattern_policy | iosxe_l2.yaml

V-220662 CISC-L2-000160

The Cisco switch must have Storm Control configured on all host-facing switchports.

Purpose and required control: Loop through all access ports to ensure multiple strings are applied

Evidence read: show running-config, show interfaces status

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "access"}}]}}

- Required on each interface: storm-control broadcast level

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: cat3 | interface_config_policy | iosxe_l2.yaml

V-220663 CISC-L2-000170

The Cisco switch must have IGMP or MLD Snooping configured on all VLANs.

Purpose and required control: Ensure IGMP and IPv6 MLD snooping are not disabled globally or for individual VLANs

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Forbidden: IGMP snooping is disabled globally or for a VLAN (read from show running-config)
- Forbidden: IPv6 MLD snooping is disabled globally or for a VLAN (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: IGMP and IPv6 MLD snooping are not disabled globally or for individual VLANs.

Configured failure explanation: IGMP or IPv6 MLD snooping is disabled globally or for one or more VLANs.

Traceability: cat3 | command_pattern_policy | iosxe_l2.yaml

V-220664 CISC-L2-000180

The Cisco switch must implement Rapid STP where VLANs span multiple switches with redundant links.

Purpose and required control: Looking for list of strings in "show run"

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Alternative: {"strings": ["spanning-tree mode rapid-pvst"], "command": "show running-config"} (read from show running-config)
- Alternative: {"strings": ["spanning-tree mode mst"], "command": "show running-config"} (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: cat2 | command_pattern_policy | iosxe_l2.yaml

V-220665 CISC-L2-000190

The Cisco switch must enable Unidirectional Link Detection (UDLD) to protect against one-way connections.

Purpose and required control: Looking for list of strings in "show run | inc udld"

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: {"command": "show running-config", "strings": ["udld enable"]} (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: cat2 | command_pattern_policy | iosxe_l2.yaml

V-220666 CISC-L2-000200

The Cisco switch must have all trunk links enabled statically.

Purpose and required control: Ensure all switchports report Negotiation of Trunking as Off

Evidence read: show interfaces switchport | include Negotiation of Trunking

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: at least one switchport reports trunk negotiation Off (read from show interfaces switchport | include Negotiation of Trunking)
- Forbidden: switchport with trunk negotiation not set to Off (read from show interfaces switchport | include Negotiation of Trunking)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: All reported switchports have Negotiation of Trunking set to Off.

Configured failure explanation: One or more switchports have Negotiation of Trunking enabled or set to an unexpected value.

Traceability: cat2 | command_pattern_policy | iosxe_l2.yaml

V-220667 CISC-L2-000210

The Cisco switch must have all disabled switch ports assigned to an unused VLAN.

Purpose and required control: Exempt 802.1X access ports; require all other disabled or disconnected access ports to be shut down and assigned to the unused VLAN

Evidence read: show running-config, show interfaces status

Site input: unused_vlan = 999. VLAN reserved for unused access ports. Ask the network owner for the approved number.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "access", "admin_state": "disabled_or_notconnect"}}]}}

- Interface exemption: 802.1X authentication configured
- Required on each interface: shutdown configured
- Required on each interface: unused VLAN {{ unused_vlan }} assigned

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: All non-exempt disabled or disconnected access ports are shut down and assigned to the unused VLAN.

Configured failure explanation: One or more non-exempt disabled or disconnected access ports are not shut down or are not assigned to the unused VLAN.

Traceability: cat2 | interface_config_policy | iosxe_l2.yaml

V-220668 CISC-L2-000220

The Cisco switch must not have the default VLAN assigned to any host-facing switch ports.

Purpose and required control: Loop through access ports to make sure none are assigned to VLAN 1

Evidence read: show running-config, show interfaces status

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "access"}}]}}

- Required on each interface: non-default access VLAN configured
- Forbidden on each interface: access VLAN 1

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: cat2 | interface_config_policy | iosxe_l2.yaml

V-220669 CISC-L2-000230

The Cisco switch must have the default VLAN pruned from all trunk ports that do not require it.

Purpose and required control: Loop through trunk ports to ensure all have VLAN 1 pruned

Evidence read: show running-config, show interfaces status

Site input: trunk_policy.additional_pruned_vlans = []. Additional VLAN numbers forbidden on trunks, where referenced by a check.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "trunk"}}]}}

- Required on each interface: explicit trunk allowed VLAN list
- Forbidden on each interface: VLAN 1 allowed on trunk
- Forbidden on each interface for profile values: profile-pruned VLAN {{ item }} allowed on trunk (profile field trunk_policy.additional_pruned_vlans)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: cat2 | interface_config_policy | iosxe_l2.yaml

V-220670 CISC-L2-000240

The Cisco switch must not use the default VLAN for management traffic.

Purpose and required control: Ensure VLAN 1 is not the management VLAN and has no IP address assigned

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile

lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Forbidden: interface Vlan1 has an IP address (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: cat2 | command_pattern_policy | iosxe_l2.yaml

V-220671 CISC-L2-000250

The Cisco switch must have all user-facing or untrusted ports configured as access switch ports.

Purpose and required control: Require switchport mode access on every physical copper Ethernet port; ignore SFP/fiber ports used as uplinks

Evidence read: show running-config, show interfaces status

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"media_type": {"regex": "BaseTX"}}}]}}

- Required on each interface: switchport mode access

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: All physical copper Ethernet ports are statically configured as access ports.

Configured failure explanation: One or more physical copper Ethernet ports are not statically configured as access ports.

Traceability: cat2 | interface_config_policy | iosxe_l2.yaml

V-220672 CISC-L2-000260

The Cisco switch must have the native VLAN assigned to an ID other than the default VLAN for all 802.1q trunk links.

Purpose and required control: Ensure every physical trunk has an explicitly configured native VLAN other than VLAN 1

Evidence read: show running-config, show interfaces status

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical", "match": {"switchport_mode": "trunk"}}]}}

- Required on each interface: non-default native VLAN configured
- Forbidden on each interface: native VLAN 1

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Automated validation matched the configured strings/patterns for this requirement.

Configured failure explanation: Automated validation found missing required strings or present forbidden strings.

Traceability: cat2 | interface_config_policy | iosxe_l2.yaml

V-220673 CISC-L2-000270

The Cisco switch must not have any switchports assigned to the native VLAN.

Purpose and required control: Ensure no physical switchport is assigned as an access port in the profile-defined native VLAN

Evidence read: show interfaces status

Site input: native_vlan = 333. Approved native VLAN number on trunks.

How it checks and verifies: The parser selects interfaces in the scope below. Global conditions apply to the full running configuration. Every selected interface must satisfy all required conditions, every alternative group, and all forbidden-value tests. A matching exemption skips checks on that interface and counts it as passing. Any failure causes the configured failure status. With no matching interfaces the check can pass unless require_candidates is true; then it returns Error. This is configuration inspection, not a traffic test.

Exact selection scope: {"interfaces": {"include": [{"type": "physical"}]}}

- require_candidates: true
- Forbidden parsed interface value: access VLAN matches native VLAN {{ native_vlan }} (profile field native_vlan)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: No physical switchports are assigned to native VLAN {{ native_vlan }}.

Configured failure explanation: One or more physical switchports are assigned to native VLAN {{ native_vlan }}.

Traceability: cat3 | interface_config_policy | iosxe_l2.yaml

IOSXE NDM checks

42 implemented checks. File: iosxe_ndm.yaml.

V-220518 CISC-ND-000010

The Cisco switch must be configured to limit the number of concurrent management sessions to an organization-defined number.

Purpose and required control: Validate session-limit under every VTY section, or count the available VTY lines when session-limit is unsupported

Evidence read: show running-config

Site input: variables.max_concurrent_management_sessions = 16. Organization-approved maximum concurrent management sessions, a positive integer.

How it checks and verifies: The maximum comes from the profile. If any VTY section has session-limit, all VTY sections must have a valid session-limit no greater than that maximum. Otherwise the app counts distinct configured VTY line numbers and compares the count to the maximum. Missing VTY sections or invalid input cannot establish compliance. This is a configured-capacity check, not a count of current logged-in users.

- command: "show running-config"

- maximum_sessions_profile_key: "max_concurrent_management_sessions"

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The number of concurrent management sessions does not exceed the organization-defined maximum of {{ max_concurrent_management_sessions }}.

Configured failure explanation: The VTY session limit or number of available VTY lines exceeds the organization-defined maximum of {{ max_concurrent_management_sessions }}.

Traceability: cat2 | vty_session_limit_policy | iosxe_ndm.yaml

V-220519 CISC-ND-000090

The Cisco switch must be configured to automatically audit account creation.

Purpose and required control: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: archive configuration section (read from show running-config)
- Required: log config subsection under archive (read from show running-config)
- Required: logging enable under archive log config (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Archive configuration auditing is enabled under the archive log config hierarchy.

Configured failure explanation: The archive section, log config subsection, or nested logging enable command is missing.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220520 CISC-ND-000100

The Cisco switch must be configured to automatically audit account modification.

Purpose and required control: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: archive configuration section (read from show running-config)
- Required: log config subsection under archive (read from show running-config)
- Required: logging enable under archive log config (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Archive configuration auditing is enabled under the archive log config hierarchy.

Configured failure explanation: The archive section, log config subsection, or nested logging enable command is missing.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220521 CISC-ND-000110

The Cisco switch must be configured to automatically audit account disabling actions.

Purpose and required control: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: archive configuration section (read from show running-config)
- Required: log config subsection under archive (read from show running-config)
- Required: logging enable under archive log config (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Archive configuration auditing is enabled under the archive log config hierarchy.

Configured failure explanation: The archive section, log config subsection, or nested logging enable command is missing.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220522 CISC-ND-000120

The Cisco switch must be configured to automatically audit account removal actions.

Purpose and required control: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: archive configuration section (read from show running-config)
- Required: log config subsection under archive (read from show running-config)
- Required: logging enable under archive log config (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Archive configuration auditing is enabled under the archive log config hierarchy.

Configured failure explanation: The archive section, log config subsection, or nested logging enable command is missing.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220523 CISC-ND-000140

The Cisco switch must be configured to enforce approved authorizations for controlling the flow of management information within the device based on control policies.

Purpose and required control: Require the profile-defined management ACL inbound on every VTY section and permit only profile-approved management networks

Evidence read: show running-config

Site input: management_access.acl_name = "MANAGEMENT_NET". Name of the ACL applied inbound to VTY management lines.

Site input: management_access.networks = [{"network_address": "192.0.2.0", "subnet_mask": "255.255.255.0"}]. Approved source networks as network_address and subnet_mask. Use an ordinary subnet mask, not an ACL wildcard mask.

How it checks and verifies: Every parsed VTY section must contain access-class followed by the profile ACL name and in (optional vrf-also). The named standard or extended ACL must contain permits for all approved source networks and no permit source outside that set. The parser converts profile subnet masks to wildcard masks for comparison. Missing VTY sections or invalid/missing profile network data returns Error. Unparseable permit sources fail. The check compares source networks; it does not prove all aspects of traffic filtering.

Decision: Criteria met: NotAFinding. Criteria not met: Not_Reviewed. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Every VTY section uses the required inbound management ACL, which permits only the profile-approved management networks.

Configured failure explanation: A VTY section is missing the required inbound management ACL, or the ACL permits an unapproved source network.

Traceability: cat2 | management_access_policy | iosxe_ndm.yaml

V-220524 CISC-ND-000150

The Cisco switch must be configured to enforce the limit of three consecutive invalid logon attempts, after which time it must lock out the user account from accessing the device for 15 minutes.

Purpose and required control: Ensure login blocking is configured for 900 seconds after three failed attempts within 120 seconds

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: login block-for 900 attempts 3 within 120 (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Login blocking is configured for 900 seconds after three failed attempts within 120 seconds.

Configured failure explanation: The required login block-for 900 attempts 3 within 120 command is missing.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220525 CISC-ND-000160

The Cisco switch must be configured to display the Standard Mandatory DoD Notice and Consent Banner before granting access to the device.

Purpose and required control: Verify the login banner contains every mandatory DoD notice and consent clause

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app extracts complete login banner blocks and normalizes case and punctuation. One complete banner must contain all of the built-in DoD notice and consent clauses. Missing clauses fail. This is a text comparison; it does not open an interactive login session to verify presentation.

- command: "show running-config"
- banner_type: "login"

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The configured login banner contains all mandatory Standard DoD Notice and Consent clauses.

Configured failure explanation: The login banner is missing or does not contain all mandatory Standard DoD Notice and Consent clauses.

Traceability: cat2 | dod_banner_policy | iosxe_ndm.yaml

V-220526 CISC-ND-000210

The Cisco device must be configured to audit all administrator activity.

Purpose and required control: Ensure administrator configuration changes generate informational logging messages

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: logging userinfo (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Administrator activity logging is enabled with logging userinfo.

Configured failure explanation: The logging userinfo command is missing.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220528 CISC-ND-000280

The Cisco switch must produce audit records containing information to establish when (date and time) the events occurred.

Purpose and required control: Ensure log messages use millisecond local date and time timestamps with timezone and year

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: service timestamps log datetime msec localtime show-timezone year (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Log messages use millisecond local date and time timestamps with timezone and year.

Configured failure explanation: The service timestamps log datetime msec localtime show-timezone year command is missing.

Exact-match limit: the final year token is required, and extra tokens after year fail this expression.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220529 CISC-ND-000290

The Cisco switch must produce audit records containing information to establish where the events occurred.

Purpose and required control: Require log-input on every deny statement in each interface-bound IPv4 ACL

Evidence read: show running-config, show ip access-lists

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The parser reads ACL entries in the scope specified below and requires selected deny entries to carry logging. Missing logging on a selected deny fails. Review the exact scope: this does not require logging on ACL entries outside it or prove that log messages reach a server.

- deny_statements: {"interface_bound_only": true, "include_standard_acls": true, "include_extended_acls": true, "include_ipv6_acls": false, "require_keyword": "log-input"}

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Every deny statement in each interface-bound IPv4 ACL includes log-input.

Configured failure explanation: One or more deny statements in an interface-bound IPv4 ACL are missing log-input.

Traceability: cat2 | acl_deny_logging_policy | iosxe_ndm.yaml

V-220530 CISC-ND-000330

The Cisco switch must be configured to generate audit records containing the full-text recording of privileged commands.

Purpose and required control: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile

lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: archive configuration section (read from show running-config)
- Required: log config subsection under archive (read from show running-config)
- Required: logging enable under archive log config (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Archive configuration auditing is enabled under the archive log config hierarchy.

Configured failure explanation: The archive section, log config subsection, or nested logging enable command is missing.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220531 CISC-ND-000380

The Cisco switch must be configured to protect audit information from unauthorized modification.

Purpose and required control: Logging persistent is not configured; temporarily Not Applicable pending future implementation and validation

Evidence read: No command; manual preset.

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The engine does not independently test the device for this item. It returns the configured fail status as a preset, even when that status is NotAFinding or Not_Applicable. The assessor must obtain the organizational or device evidence described below and make the final decision.

Decision and required manual verification: Current preset: Not_Applicable. Logging persistent is not configured.

Traceability: cat2 | manual_review | iosxe_ndm.yaml

V-220532 CISC-ND-000390

The Cisco switch must be configured to protect audit information from unauthorized deletion.

Purpose and required control: Logging persistent is not configured; temporarily Not Applicable pending future implementation and validation

Evidence read: No command; manual preset.

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The engine does not independently test the device for this item. It returns the configured fail status as a preset, even when that status is NotAFinding or Not_Applicable. The assessor must obtain the organizational or device evidence described below and make the final decision.

Decision and required manual verification: Current preset: Not_Applicable. Logging persistent is not configured.

Traceability: cat2 | manual_review | iosxe_ndm.yaml

V-220533 CISC-ND-000460

The Cisco switch must be configured to limit privileges to change the software resident within software libraries.

Purpose and required control: Logging persistent is not configured; temporarily Not Applicable pending future implementation and validation

Evidence read: No command; manual preset.

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The engine does not independently test the device for this item. It returns the configured fail status as a preset, even when that status is NotAFinding or Not_Applicable. The assessor must obtain the organizational or device evidence described below and make the final decision.

Decision and required manual verification: Current preset: Not_Applicable. Logging persistent is not configured.

Traceability: cat2 | manual_review | iosxe_ndm.yaml

V-220534 CISC-ND-000470

The Cisco switch must be configured to prohibit the use of all unnecessary and nonsecure functions and services.

Purpose and required control: Ensure prohibited and unnecessary services are not enabled in the running configuration

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Forbidden: prohibited or unnecessary service is enabled (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: None of the prohibited or unnecessary services are enabled.

Configured failure explanation: One or more prohibited or unnecessary services are enabled.

Traceability: cat1 | command_pattern_policy | iosxe_ndm.yaml

V-220535 CISC-ND-000490

The Cisco switch must be configured with only one local account to be used as the account of last resort in the event the authentication server is unavailable.

Purpose and required control: Require exactly one local username using the profile-defined common-criteria password policy and require the profile-defined TACACS group before local in the default login authentication order

Evidence read: show running-config

Site input: variables.local_password_policy = "PASSWORD_POLICY". Name of the aaa common-criteria policy referenced by local user configuration.

Site input: variables.tacacs_group = "ISE-TACACS". TACACS+ server group name used by the default login method list.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: one local username account is configured (read from show running-config)
- Forbidden: multiple local username accounts are configured (read from show running-config)
- Required for every profile value: local username using common-criteria policy {{ item }} (profile field local_password_policy; read from show running-config)
- Required for every profile value: default login authentication uses group {{ item }} before local (profile field tacacs_group; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Exactly one local account uses the required common-criteria password policy, and the required TACACS group precedes local in the default login authentication order.

Configured failure explanation: The local-account count, common-criteria password policy, or default AAA login authentication order is incorrect.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220537 CISC-ND-000550

The Cisco switch must be configured to enforce a minimum 15-character password length.

Purpose and required control: Require min-length 15 under the profile-defined AAA common-criteria password policy

Evidence read: show running-config

Site input: variables.local_password_policy = "PASSWORD_POLICY". Name of the aaa common-criteria policy referenced by local user configuration.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required for every profile value: min-length 15 under common-criteria policy {{ item }} (profile field local_password_policy; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The required minimum password length is configured under the profile-defined common-criteria policy.

Configured failure explanation: The profile-defined common-criteria policy is missing min-length 15.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220538 CISC-ND-000570

The Cisco switch must be configured to enforce password complexity by requiring that at least one uppercase character be used.

Purpose and required control: Require upper-case 1 under the profile-defined AAA common-criteria password policy

Evidence read: show running-config

Site input: variables.local_password_policy = "PASSWORD_POLICY". Name of the aaa common-criteria policy referenced by local user configuration.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required for every profile value: upper-case 1 under common-criteria policy {{ item }} (profile field local_password_policy; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The uppercase-character requirement is configured under the profile-defined common-criteria policy.

Configured failure explanation: The profile-defined common-criteria policy is missing upper-case 1.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220539 CISC-ND-000580

The Cisco switch must be configured to enforce password complexity by requiring that at least one lowercase character be used.

Purpose and required control: Require lower-case 1 under the profile-defined AAA common-criteria password policy

Evidence read: show running-config

Site input: variables.local_password_policy = "PASSWORD_POLICY". Name of the aaa common-criteria policy referenced by local user configuration.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required for every profile value: lower-case 1 under common-criteria policy {{ item }} (profile field local_password_policy; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The lowercase-character requirement is configured under the profile-defined common-criteria policy.

Configured failure explanation: The profile-defined common-criteria policy is missing lower-case 1.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220540 CISC-ND-000590

The Cisco switch must be configured to enforce password complexity by requiring that at least one numeric character be used.

Purpose and required control: Require numeric-count 1 under the profile-defined AAA common-criteria password policy

Evidence read: show running-config

Site input: variables.local_password_policy = "PASSWORD_POLICY". Name of the aaa common-criteria policy referenced by local user configuration.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required for every profile value: numeric-count 1 under common-criteria policy {{ item }} (profile field local_password_policy; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The numeric-character requirement is configured under the profile-defined common-criteria policy.

Configured failure explanation: The profile-defined common-criteria policy is missing numeric-count 1.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220541 CISC-ND-000600

The Cisco switch must be configured to enforce password complexity by requiring that at least one special character be used.

Purpose and required control: Require special-case 1 under the profile-defined AAA common-criteria password policy

Evidence read: show running-config

Site input: variables.local_password_policy = "PASSWORD_POLICY". Name of the aaa common-criteria policy referenced by local user configuration.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required for every profile value: special-case 1 under common-criteria policy {{ item }} (profile field local_password_policy; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The special-character requirement is configured under the profile-defined common-criteria policy.

Configured failure explanation: The profile-defined common-criteria policy is missing special-case 1.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220542 CISC-ND-000610

The Cisco switch must be configured to require that when a password is changed, the characters are changed in at least eight of the positions within the password.

Purpose and required control: Require char-changes 8 under the profile-defined AAA common-criteria password policy

Evidence read: show running-config

Site input: variables.local_password_policy = "PASSWORD_POLICY". Name of the aaa common-criteria policy referenced by local user configuration.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required for every profile value: char-changes 8 under common-criteria policy {{ item }} (profile field local_password_policy; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The required password-position changes are configured under the profile-defined common-criteria policy.

Configured failure explanation: The profile-defined common-criteria policy is missing char-changes 8.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220543 CISC-ND-000620

The Cisco switch must only store cryptographic representations of passwords.

Purpose and required control: Ensure service password-encryption is enabled

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: service password-encryption enabled (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Service password encryption is enabled.

Configured failure explanation: The service password-encryption command is missing.

Traceability: cat1 | command_pattern_policy | iosxe_ndm.yaml

V-220544 CISC-ND-000720

The Cisco switch must be configured to terminate all network connections associated with device management after five minutes of inactivity.

Purpose and required control: Require session-timeout 5 or exec-timeout 5 0 under every VTY line section

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: at least one VTY line section is configured (read from show running-config)
- Forbidden: VTY line section missing an approved five-minute inactivity timeout (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Every VTY line section has a five-minute inactivity timeout configured.

Configured failure explanation: One or more VTY line sections are missing session-timeout 5 or exec-timeout 5 0.

Traceability: cat1 | command_pattern_policy | iosxe_ndm.yaml

V-220545 CISC-ND-000880

The Cisco switch must be configured to automatically audit account enabling actions.

Purpose and required control: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: archive configuration section (read from show running-config)
- Required: log config subsection under archive (read from show running-config)
- Required: logging enable under archive log config (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Archive configuration auditing is enabled under the archive log config hierarchy.

Configured failure explanation: The archive section, log config subsection, or nested logging enable command is missing.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220547 CISC-ND-000980

The Cisco switch must be configured to allocate audit record storage capacity in accordance with organization-defined audit record storage requirements.

Purpose and required control: Ensure logging buffered specifies a numeric buffer size and a valid logging severity level

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: logging buffered with numeric size and valid severity level (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Logging buffered is configured with a numeric buffer size and valid severity level.

Configured failure explanation: A valid logging buffered size and severity level are not configured.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220548 CISC-ND-001000

The Cisco switch must be configured to generate an alert for all audit failure events.

Purpose and required control: Ensure remote logging uses a valid named trap severity level

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: logging trap with valid named severity level (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Remote logging is configured with a valid named trap severity level.

Configured failure explanation: A valid named logging trap severity level is not configured.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220549 CISC-ND-001030

The Cisco switch must be configured to synchronize its clock with the primary and secondary time sources using redundant authoritative time sources.

Purpose and required control: Require at least two NTP server commands and every NTP server defined in the selected profile

Evidence read: show running-config

Site input: variables.ntp_servers = ["192.0.2.22", "192.0.2.23"]. Exact IP addresses or hostnames following ntp server. List at least two approved sources.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: at least two NTP server commands (read from show running-config)
- Required for every profile value: profile-defined NTP server {{ item }} (profile field ntp_servers; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: At least two profile-defined authoritative NTP servers are configured.

Configured failure explanation: Fewer than two NTP servers are configured or a profile-defined NTP server is missing.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220552 CISC-ND-001130

The Cisco switch must be configured to authenticate SNMP messages using a FIPS-validated Keyed-Hash Message Authentication Code (HMAC).

Purpose and required control: Require at least one SNMPv3 user using SHA or a supported SHA-2 authentication protocol

Evidence read: show snmp user

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: FIPS-acceptable SNMPv3 HMAC authentication protocol (read from show snmp user)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: At least one SNMPv3 user uses a FIPS-acceptable SHA-based HMAC authentication protocol.

Configured failure explanation: No SNMPv3 user using an approved SHA-based HMAC authentication protocol was found.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220553 CISC-ND-001140

The Cisco switch must be configured to encrypt SNMP messages using a FIPS 140-2 approved algorithm.

Purpose and required control: Require at least one SNMPv3 user using AES-128, AES-192, or AES-256 privacy

Evidence read: show snmp user

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: FIPS-approved AES SNMPv3 privacy protocol (read from show snmp user)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: At least one SNMPv3 user uses a FIPS-approved AES privacy protocol.

Configured failure explanation: No SNMPv3 user using AES-128, AES-192, or AES-256 privacy was found.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220554 CISC-ND-001150

The Cisco switch must be configured to authenticate Network Time Protocol (NTP) sources using authentication that is cryptographically based.

Purpose and required control: Require every profile-defined NTP server to use a trusted HMAC-SHA2-256 authentication key

Evidence read: show running-config

Site input: variables.ntp_servers = ["192.0.2.22", "192.0.2.23"]. Exact IP addresses or hostnames following ntp server. List at least two approved sources.

How it checks and verifies: At least the configured minimum number of unique profile NTP servers is required. The app requires global ntp authenticate. Each profile server must have an ntp server association using a configured key, that key must be trusted, and its algorithm must be in approved_algorithms below. Missing authentication, association, trust, or allowed algorithm fails. Invalid profile server/algorithm input returns Error. This checks configuration, not successful time synchronization or independent cryptographic validation.

- command: "show running-config"
- ntp_servers_profile_key: "ntp_servers"
- minimum_servers: 2
- approved_algorithms: ["hmac-sha2-256", "md5"]

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Every profile-defined NTP server uses a trusted HMAC-SHA2-256 authentication key and global NTP authentication is enabled.

Configured failure explanation: Global NTP authentication is disabled, or one or more profile-defined NTP servers do not use a trusted HMAC-SHA2-256 key.

The algorithm setting above accepts MD5 despite the narrower configured pass/fail wording. Treat the algorithm list as the implemented logic.

Traceability: cat2 | ntp_authentication_policy | iosxe_ndm.yaml

V-220555 CISC-ND-001200

The Cisco switch must be configured to use FIPS-validated Keyed-Hash Message Authentication Code (HMAC) to protect the integrity of remote maintenance sessions.

Purpose and required control: Require the approved SSH HMAC algorithms in the configured order

Evidence read: show running-config | include ssh

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: approved SSH HMAC algorithm list (read from show running-config | include ssh)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: SSH uses the approved HMAC-SHA2 integrity algorithms in the required order.

Configured failure explanation: The required SSH HMAC algorithm list is missing or incorrect.

Traceability: cat1 | command_pattern_policy | iosxe_ndm.yaml

V-220556 CISC-ND-001210

The Cisco switch must be configured to implement cryptographic mechanisms to protect the confidentiality of remote maintenance sessions.

Purpose and required control: Require the approved SSH AES-CTR encryption algorithms in the configured order

Evidence read: show running-config | include ssh

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: approved SSH AES-CTR encryption algorithm list (read from show running-config | include ssh)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: SSH uses the approved AES-CTR encryption algorithms in the required order.

Configured failure explanation: The required SSH AES-CTR encryption algorithm list is missing or incorrect.

Traceability: cat1 | command_pattern_policy | iosxe_ndm.yaml

V-220559 CISC-ND-001250

The Cisco switch must be configured to generate log records when administrator privileges are deleted.

Purpose and required control: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: archive configuration section (read from show running-config)
- Required: log config subsection under archive (read from show running-config)
- Required: logging enable under archive log config (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Archive configuration auditing is enabled under the archive log config hierarchy.

Configured failure explanation: The archive section, log config subsection, or nested logging enable command is missing.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220560 CISC-ND-001260

The Cisco switch must be configured to generate audit records when successful/unsuccessful logon attempts occur.

Purpose and required control: Ensure successful and failed login attempts are logged

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: login on-success log (read from show running-config)
- Required: login on-failure log (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Successful and failed login attempts are configured for logging.

Configured failure explanation: Login success logging, login failure logging, or both are missing.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220561 CISC-ND-001270

The Cisco switch must be configured to generate log records for privileged activities.

Purpose and required control: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

Evidence read: show running-config

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: archive configuration section (read from show running-config)
- Required: log config subsection under archive (read from show running-config)
- Required: logging enable under archive log config (read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: Archive configuration auditing is enabled under the archive log config hierarchy.

Configured failure explanation: The archive section, log config subsection, or nested logging enable command is missing.

Traceability: cat2 | command_pattern_policy | iosxe_ndm.yaml

V-220565 CISC-ND-001370

The Cisco switch must be configured to use at least two authentication servers for the purpose of authenticating users prior to granting administrative access.

Purpose and required control: Require at least two profile-defined RADIUS server blocks with the correct address, authentication port, accounting port, key, and AAA group membership

Evidence read: show running-config

Site input: endpoint_authentication.radius_group = "ISE-RADIUS". AAA RADIUS server group name as configured on the switch.

Site input: endpoint_authentication.radius_server_addresses = {"RADIUS-SERVER-1": "192.0.2.18", "RADIUS-SERVER-2": "192.0.2.19"}. Mapping of each listed RADIUS configuration name to its expected IPv4 address.

Site input: endpoint_authentication.radius_servers = ["RADIUS-SERVER-1", "RADIUS-SERVER-2"]. List of required radius server names; use the configuration names, not necessarily DNS hostnames.

How it checks and verifies: The profile must define at least two unique RADIUS server names and an IPv4 mapping for each. Each corresponding radius server section must use the mapped address, authentication port 1812, accounting port 1813, and a key command. The named AAA RADIUS group must contain each required server name. Missing or mismatched configuration fails. The app does not contact the RADIUS servers or test a login.

- radius_group_profile_key: "endpoint_authentication.radius_group"
- radius_servers_profile_key: "endpoint_authentication.radius_servers"
- radius_addresses_profile_key: "endpoint_authentication.radius_server_addresses"
- auth_port: 1812
- acct_port: 1813

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: At least two profile-defined RADIUS servers have the required addresses, ports, keys, and AAA group membership.

Configured failure explanation: One or more required RADIUS server definitions, addresses, ports, keys, or AAA group memberships are missing or incorrect.

Traceability: cat1 | radius_server_policy | iosxe_ndm.yaml

V-220566 CISC-ND-001410

The Cisco switch must be configured to support organizational requirements to conduct backups of the configuration when changes occur.

Purpose and required control: Temporarily mark NotAFinding because configuration backups are performed through Cisco Catalyst Center

Evidence read: No command; manual preset.

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The engine does not independently test the device for this item. It returns the configured fail status as a preset, even when that status is NotAFinding or Not_Applicable. The assessor must obtain the organizational or device evidence described below and make the final decision.

Decision and required manual verification: Current preset: NotAFinding. Configuration backups are performed through Cisco Catalyst Center.

Traceability: cat2 | manual_review | iosxe_ndm.yaml

V-220567 CISC-ND-001440

The Cisco switch must be configured to obtain its public key certificates from an appropriate certificate policy through an approved service provider.

Purpose and required control: Not Applicable because these access switches do not use organization-managed public key certificates

Evidence read: No command; manual preset.

Site input: No check-specific profile input. Supply the connection inputs for a live scan. For a manual entry, obtain and document the organizational evidence described in the preset explanation.

How it checks and verifies: The engine does not independently test the device for this item. It returns the configured fail status as a preset, even when that status is NotAFinding or Not_Applicable. The assessor must obtain the organizational or device evidence described below and make the final decision.

Decision and required manual verification: Current preset: Not_Applicable. Not Applicable. The switch does not use organization-managed public key certificates. Only Cisco factory-installed/default certificates and trustpoints are present. No locally configured CA enrollment trustpoint or certificate-based management service is used.

Traceability: cat2 | manual_review | iosxe_ndm.yaml

V-220568 CISC-ND-001450

The Cisco switch must be configured to send log data to at least two central log servers for the purpose of forwarding alerts to the administrators and the information system security officer (ISSO).

Purpose and required control: Require at least two logging host commands and every logging server defined in the selected profile

Evidence read: show running-config

Site input: variables.logging_servers = ["192.0.2.5", "192.0.2.6"]. Exact IP addresses or hostnames following logging host in the configuration. List at least two approved destinations.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Required: at least two logging host commands (read from show running-config)
- Required for every profile value: profile-defined logging server {{ item }} (profile field logging_servers; read from show running-config)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: At least two profile-defined central logging servers are configured.

Configured failure explanation: Fewer than two logging hosts are configured or a profile-defined logging server is missing.

Traceability: cat1 | command_pattern_policy | iosxe_ndm.yaml

V-220569 CISC-ND-001470

The Cisco switch must be running an IOS release that is currently supported by Cisco Systems.

Purpose and required control: Match the installed IOS-XE release against the profile-defined list of approved Cisco-supported versions

Evidence read: show version

Site input: variables.supported_iosxe_versions = ["17.12.08"]. Exact approved IOS-XE release strings from show version. The shipped value is 17.12.08; the organization must maintain the list.

How it checks and verifies: The app searches the named command output. Every Required condition must match; at least one Alternative must match when alternatives exist; no Forbidden condition may match. Profile lists expand into one condition per value. Empty criteria produce Not_Reviewed. Most searches ignore case. Exact expressions are recorded with this package in check_definitions.json and the YAML.

- Alternative from profile: profile-approved IOS-XE version {{ item }} (profile field supported_iosxe_versions; read from show version)

Decision: Criteria met: NotAFinding. Criteria not met: Open. Missing output or evaluation error: Error. Evaluator exceptions described above take priority.

Configured pass explanation: The installed IOS-XE release matches a profile-defined version confirmed to be supported by Cisco.

Configured failure explanation: The installed IOS-XE release is not in the profile-defined list of Cisco-supported versions.

Traceability: cat1 | command_pattern_policy | iosxe_ndm.yaml